

Dimension-saturating local tomography from Fourier-cat dark events

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An exact dark event is a sensitive null test for a linear-optical device, but its unprobed probability is quadratic in a small coherent error and has no regular local inverse. We show that positive and negative calibrated output probes turn dark-event counts into linear phase-sensitive observables. A general signed-probe theorem gives their limiting Jacobian and proves that one dark event contributes at most two real rows. Number-state inputs face an exact obstruction: in an m -mode Fourier interferometer, an $(m-1)$ -dimensional subspace of off-diagonal output generators is indistinguishable from input phases, even after an arbitrary known output analyzer. A coherent Fourier-cat input removes this continuous gauge. For every $m \geq 2$ and every $n > 2$ divisible by m , we construct $m(m-1)/2$ dark outcomes and two fixed probes whose $m(m-1)$ signed contrasts identify every off-diagonal Hermitian error locally. Within the weak-probe, regular-local protocol defined here, both the outcome and scalar-contrast counts attain their dimension lower bounds; four programmed settings suffice for every m , although we do not prove that four is the minimum possible number of settings. At $m = n = 4$ we additionally give a sparse exact implementation with Jacobian determinant $-177147/256$. We prove an $O(\epsilon^2)$ finite-probe bias bound and show how internal coherent errors propagate to the estimated output generator. The Fourier-cat suppression rule and displaced-null principle are known; the contribution is their explicit bosonic identifiability construction together with the Fock gauge theorem and exact dimension-saturating certificates.

I. INTRODUCTION

Destructive multiphoton interference produces output events with exactly zero ideal probability. Fourier suppression laws generalize the Hong–Ou–Mandel effect to multiport devices [1–5], and forbidden-event populations have become practical witnesses of interference and device quality [2, 6, 7]. A scalar forbidden-event fraction, however, does not specify which coherent device error produced the counts.

There is a more basic local difficulty. If an event is dark at a nominal device, its probability has zero derivative with respect to every coherent error coordinate. The leading quadratic response cannot distinguish opposite errors. The situation parallels the non-identifiability of exact null measurements in quantum statistics. Girotti, Godley, and Guță showed that a deliberate displacement restores identifiability and can support optimal estimation in general pure-state models [8]. Here we develop a finite-mode, photon-counting implementation of that principle: a known signed optical probe supplies a reference amplitude against which unknown leakage interferes.

The central issue is not merely sensitivity but *identifiability*. Number-state preparation and number-resolved detection possess independent input and output phase gauges, familiar from linear-optical characterization [9]. At a Fourier device, the transported input-phase gauge lies entirely inside the nominally off-diagonal output-error coordinates. We prove that number-state experiments cannot see these directions even if an arbitrary

calibrated analyzer is appended. An exact four-photon search then saturates, rather than evades, this rank bound.

The missing resource is an input phase reference. We use equal superpositions of all-bunched number states. Fourier multiports and multimode cat states have a substantial prior literature [10]; the associated modular dark sectors also follow from the general symmetry suppression framework [4]. We use this known selection rule to produce a dimension-saturating frame of simultaneously dark outcomes for arbitrary mode number. General near-identity qudit-unitary estimation has also been studied with a controlled-qudit architecture [11]; our construction instead uses bosonic Fourier interference, number-resolved dark outcomes, and two fixed global probes.

Our contributions are:

1. a general count-level signed-probe theorem, including the two-real-row bound per dark event and a rigorous $O(\epsilon^2)$ finite-angle bias bound;
2. an exact $(m-1)$ -dimensional number-state gauge no-go theorem;
3. an outcome- and contrast-minimal rank-nine F_4 certificate, within the same regular-local framework, using one four-photon Fock input;
4. for all $m \geq 2$ and $n > 2$ divisible by m , a dimension-saturating Fourier-cat construction identifying all $m(m-1)$ off-diagonal Hermitian coordinates from $m(m-1)/2$ dark outcomes and four programmed signed settings;
5. a sparse exact F_4 implementation with a rational full-rank certificate;

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6. a first-order adjoint rule connecting arbitrary internal coherent circuit errors to the reconstructed output generator.

The result is local tomography on the declared $m(m-1)$ -dimensional off-diagonal slice. It is not global unitary tomography and does not identify the $m-1$ output-diagonal $SU(m)$ generators.

II. MODEL AND PARAMETER TARGET

Let U_0 be an m -mode passive interferometer and $\hat{U}_0$ its action on a fixed n -photon sector. For number occupations $\mathbf{r}, \mathbf{s}$, the normalized transition amplitude is

$$\mathcal{A}_{\mathbf{r}, \mathbf{s}}(U) = \frac{\text{per } U[\mathbf{s}, \mathbf{r}]}{\sqrt{\prod_j r_j! \prod_k s_k!}}. \quad (1)$$

Rows and columns are repeated according to $\mathbf{s}$ and $\mathbf{r}$. The unknown coherent output generator belongs to

$$\begin{aligned} \mathcal{H}_{\text{off}}(m) &= \{H = H^\dagger : H_{aa} = 0\}, \\ \dim_{\mathbb{R}} \mathcal{H}_{\text{off}}(m) &= m(m-1). \end{aligned} \quad (2)$$

For the explicit four-mode certificates we use

$$(F_4)_{kj} = \frac{i^{kj}}{2}, \quad 0 \leq k, j < 4. \quad (3)$$

The twelve off-diagonal Hermitian generators are

$$\begin{aligned} X_{pq} &= |p\rangle\langle q| + |q\rangle\langle p|, \\ Y_{pq} &= -i|p\rangle\langle q| + i|q\rangle\langle p|, \quad 0 \leq p < q < 4. \end{aligned} \quad (4)$$

We fix the coordinate order

$$\boldsymbol{\theta} = (x_{01}, y_{01}, x_{02}, y_{02}, x_{03}, y_{03}, x_{12}, y_{12}, x_{13}, y_{13}, x_{23}, y_{23}) \quad (5)$$

and write $G(\boldsymbol{\theta}) = \sum_{\mu} \theta_{\mu} G_{\mu}$. The unknown device is locally

$$U(\boldsymbol{\theta}) = e^{iG(\boldsymbol{\theta})} F_4. \quad (6)$$

The normalization $\text{Tr } X_{pq}^2 = \text{Tr } Y_{pq}^2 = 2$ fixes the meaning of the reported coordinates and condition numbers.

The three traceless output-diagonal generators are deliberately excluded. For an output occupation $\mathbf{s}$, a diagonal output unitary multiplies its amplitude by a phase, leaving its number probability unchanged. Moreover, at a dark event

$$\langle \mathbf{s} | i\hat{Z} \hat{U}_0 | \psi \rangle = iz(\mathbf{s}) \langle \mathbf{s} | \hat{U}_0 | \psi \rangle = 0. \quad (7)$$

They therefore do not occur in the leading dark-event Jacobian. A finite noncommuting analyzer can create higher-order sensitivity to them, but that is a different parameter model.

III. SIGNED-PROBE IDENTIFIABILITY

Consider events $e = (|\psi_e\rangle, |\mathbf{s}_e\rangle)$ that are dark at U_0 :

$$\langle \mathbf{s}_e | \hat{U}_0 | \psi_e \rangle = 0. \quad (8)$$

For a real generator coordinate $\mathbf{u}$, define the complex amplitude functional

$$\ell_e \mathbf{u} = \langle \mathbf{s}_e | i\hat{G}(\mathbf{u}) \hat{U}_0 | \psi_e \rangle. \quad (9)$$

A. Why an undisplaced null is insufficient

Without a probe,

$$p_e(\boldsymbol{\theta}) = |\ell_e \boldsymbol{\theta}|^2 + O(\|\boldsymbol{\theta}\|^3). \quad (10)$$

Thus $Dp_e(0) = 0$. More strongly, no collection of such unprobed dark probabilities has a locally Lipschitz left inverse on an open parameter neighborhood: along $\boldsymbol{\theta} = t\mathbf{u}$, a Lipschitz inverse would imply $|t|\|\mathbf{u}\| = O(t^2)$, a contradiction. This does not exclude a nonregular higher-order inverse on a restricted domain; it rules out the regular local tomography needed for a stable signed estimate.

B. Calibrated central contrast

Append a known probe $H(\mathbf{h})$ after the unknown error and program both signs:

$$p_{e, \mathbf{h}}^{\sigma}(\boldsymbol{\theta}, \epsilon) = \left| \langle \mathbf{s}_e | e^{i\sigma\epsilon\hat{H}(\mathbf{h})} e^{i\hat{G}(\boldsymbol{\theta})} \hat{U}_0 | \psi_e \rangle \right|^2, \quad \sigma = \pm 1. \quad (11)$$

We use the baseline-subtracted response

$$\begin{aligned} R_{e, \mathbf{h}}(\boldsymbol{\theta}, \epsilon) &= \epsilon^{-1} [p_{e, \mathbf{h}}^+(\boldsymbol{\theta}, \epsilon) - p_{e, \mathbf{h}}^-(\boldsymbol{\theta}, \epsilon) \\ &\quad - p_{e, \mathbf{h}}^+(0, \epsilon) + p_{e, \mathbf{h}}^-(0, \epsilon)]. \end{aligned} \quad (12)$$

This is four times the convention $[p^+ - p^-]/(4\epsilon)$ often used for a central difference.

Theorem 1 (signed-probe Jacobian). For every finite-dimensional photon sector,

$$\left. \frac{\partial}{\partial \theta_{\mu}} \lim_{\epsilon \rightarrow 0} R_{e, \mathbf{h}}(\boldsymbol{\theta}, \epsilon) \right|_{\boldsymbol{\theta}=0} = 4 \text{Re}[(\ell_e \mathbf{h})^* (\ell_e \mathbf{e}_{\mu})]. \quad (13)$$

At finite angle the derivative differs from Eq. (13) by $O(\epsilon^2)$.

A proof and a norm-explicit remainder bound are given in Appendix A. Equation (13) is a statement about observable probability differences, not an assumption that complex transition amplitudes can be measured directly. The probe amplitude $\ell_e \mathbf{h}$ acts as a phase reference.

Corollary 1 (two-row event bound). For one dark event, all limiting signed-probe rows lie in

$$\text{span}_{\mathbb{R}}\{\text{Re } \ell_e, \text{Im } \ell_e\}. \quad (14)$$

Hence one event contributes at most two independent real first-order measurements, no matter how many probes are applied.

If a limiting $q \times d$ response matrix has full column rank, some $d \times d$ row minor is nonsingular. Continuity preserves that minor for all sufficiently small nonzero calibrated ϵ . The ordinary local inverse-function theorem applied to the selected d responses then gives a local differentiable left inverse. This argument must not be confused with the algebraic Jacobian conjecture, which concerns a polynomial self-map with constant nonzero determinant and a *global polynomial* inverse. Nor does full rank at one point imply global injectivity. Our map need not be polynomial, the determinant is evaluated locally, and only local identifiability is claimed.

IV. THE NUMBER-STATE GAUGE AND ITS OPTIMAL RANK

Theorem 2 (transported input-phase gauge). Let F_m be the m -mode Fourier matrix. For every real traceless diagonal D , the generator

$$K_D = F_m D F_m^\dagger \quad (15)$$

has zero diagonal and is therefore an off-diagonal Hermitian generator. Nevertheless, for every input number state $|\mathbf{r}\rangle$,

$$e^{i\hat{K}_D} \hat{F}_m |\mathbf{r}\rangle = \hat{F}_m e^{i\hat{D}} |\mathbf{r}\rangle = e^{i\sum_j d_j r_j} \hat{F}_m |\mathbf{r}\rangle. \quad (16)$$

Consequently K_D is invisible to all output number probabilities, even after an arbitrary known output analyzer. The invisible subspace has dimension $m - 1$.

The proof is included in Appendix B. At $m = 4$, an explicit basis inside the coordinate order (5) is

$$\begin{aligned} K_A &= X_{01} + X_{03} + X_{12} + X_{23}, \\ K_B &= Y_{01} - Y_{03} + Y_{12} + Y_{23}, \\ K_C &= X_{02} + X_{13}. \end{aligned} \quad (17)$$

Thus Fock-input experiments can identify at most nine of the twelve off-diagonal coordinates.

A. Exact four-photon saturation

The bound is tight. Use input $\mathbf{r} = (1, 1, 1, 1)$ and the nine output-probe contrasts in Table I. All five outputs are exactly dark by the cyclic Fourier selection rule [1, 4].

The normalized 9×12 limiting matrix is displayed and derived in Appendix B. It annihilates exactly the three

TABLE I. An exact rank-nine design. Multiple probes on one output give separate scalar contrasts.

output $\mathbf{s}$	probes
(0, 0, 1, 3)	X_{03}, Y_{03}
(0, 0, 2, 2)	X_{02}, Y_{02}
(0, 0, 3, 1)	X_{12}, Y_{12}
(0, 1, 1, 2)	X_{23}, Y_{23}
(0, 2, 2, 0)	Y_{02}

vectors in Eq. (17); a 9×9 minor has determinant $81/8$. Therefore its rank is nine. Nine scalar contrasts are necessary for a nine-dimensional quotient, and five events are necessary by Corollary 1. The design attains both lower bounds.

V. A COHERENT PHASE REFERENCE

A. Known Fourier-cat dark sectors

Let

$$F_m(k, j) = m^{-1/2} \omega^{kj}, \quad \omega = e^{2\pi i/m}, \quad (18)$$

and define the phase-twisted all-bunched cat

$$|\Psi_\ell^{(m,n)}\rangle = \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \omega^{-\ell j} |n e_j\rangle. \quad (19)$$

For an output occupation $\mathbf{s}$ of total size n , put $Q(\mathbf{s}) = \sum_k k s_k \pmod{m}$. Direct evaluation gives

$$\mathcal{A}_{\Psi_\ell, \mathbf{s}}(F_m) = \sqrt{\frac{n!}{\prod_k s_k!}} m^{(1-n)/2} \delta_{Q(\mathbf{s}), \ell}. \quad (20)$$

Thus all outputs outside one modular sector are dark. The Fourier multiport-cat connection is known from Vourdas and Dunningham [10], and Eq. (20) is also an immediate cyclic-symmetry selection rule [4]. We recall it to specify our phase convention and resource; we do not claim it as a new suppression law. A self-contained derivation is given in Appendix C.

B. An all-mode dimension-saturating frame

We now specialize to the charge-zero cat, take $n > 2$ divisible by m , and write

$$\alpha_{\mathbf{s}} = \sqrt{\frac{n!}{\prod_a s_a!}} m^{(1-n)/2}. \quad (21)$$

For an output with $Q(\mathbf{s}) = c \neq 0$, define the phase-free dark-amplitude functional

$$L_{\mathbf{s}}(H) = \alpha_{\mathbf{s}} \sum_{a=0}^{m-1} s_a H_{a, a-c}, \quad (22)$$

where indices are modulo m . Its physical amplitude derivative is $\ell_s(H) = iL_s(H)$.

Theorem 3 (all-mode Fourier-cat frame). For every $m \geq 2$ and every $n > 2$ divisible by m , there are $m(m-1)/2$ dark outcomes and two fixed Hermitian probes R, T for which the two limiting signed contrasts per outcome have full rank on $\mathcal{H}_{\text{off}}(m)$. Thus four settings, $\pm\epsilon R$ and $\pm\epsilon T$, suffice for local identification at all sufficiently small calibrated nonzero ϵ .

Here is the explicit construction. Choose the cyclic separations $c = 1, \dots, \lfloor m/2 \rfloor$. If $2c \not\equiv 0 \pmod{m}$, use all m outcomes

$$s_{p,c} = (n-1)e_p + e_{p+c}, \quad 0 \leq p < m. \quad (23)$$

If m is even and $c = m/2$, retain only $0 \leq p < m/2$. Their charge is $np + c \equiv c$, so all are dark. Their total number is $m(m-1)/2$.

Let R be the real complete-graph probe,

$$R_{ab} = 1 \ (a \neq b), \quad R_{aa} = 0. \quad (24)$$

Orient every edge of cyclic separation $c < m/2$ by

$$T_{a,a-c} = i, \quad T_{a-c,a} = -i. \quad (25)$$

For even m , orient the remaining antipodal edges as $T_{p,p+m/2} = i$ for $0 \leq p < m/2$, with the reverse entries fixed by Hermiticity. Then

$$\begin{aligned} L_{p,c}(R) &= \alpha n, & L_{p,c}(T) &= i\alpha n & (2c \not\equiv 0), \\ L_{p,c}(R) &= \alpha n, & L_{p,c}(T) &= i\alpha(n-2) & (2c \equiv 0), \end{aligned} \quad (26)$$

where the selected outcomes share $\alpha = \sqrt{n} m^{(1-n)/2}$. The two probes therefore read the real and imaginary parts of every selected complex functional.

For completeness, the underlying block inversion is especially simple. For a non-antipodal separation, set $z_a = H_{a,a-c}$. Equation (22) becomes

$$\frac{L_{p,c}(H)}{\alpha} = (n-1)z_p + z_{p+c}. \quad (27)$$

The map is $((n-1)I + P_c)z$ with P_c a permutation. A vector in its kernel would obey $(n-1)\|z\| = \|z\|$, and hence vanishes because $n > 2$. For the antipodal class, with $z_p = x_p + iy_p$,

$$\frac{L_{p,c}(H)}{\alpha} = (n-1)z_p + z_p^* = nx_p + i(n-2)y_p, \quad (28)$$

which is also invertible. The cyclic edge classes partition all off-diagonal Hermitian entries, proving the theorem. A detailed derivation is in Appendix C.

One dark event supplies at most two real rows by Corollary 1. Consequently the theorem's $m(m-1)/2$ events and $m(m-1)$ scalar contrasts attain the corresponding lower bounds *within this weak-probe regular-local framework*. This does not prove global injectivity, nor does it prove that four programmed settings are necessary

among more general protocols. After division by the known reference magnitudes in Eq. (26), each cyclic block has condition number at most $n/(n-2)$. Without that row calibration, the worst antipodal gain ratio is at most $[n/(n-2)]^2$.

For $m \geq 3$, the smallest member of the family is $n = m$; for $m = 2$ the first allowed choice is $n = 4$. At $n = m$, the common squared amplitude normalization is

$$\alpha^2 = m^{2-m}. \quad (29)$$

This exponential-in- m factor is a probability-rate cost, not a rank or conditioning failure. With the dense unnormalized probes above, the probe-only dark-event probability scales as $p_{\text{ref}} = O(\epsilon^2 m^{4-m})$ at $n = m$; if the probe operators are normalized to fixed spectral size it scales instead as $O(\epsilon^2 m^{2-m})$. These estimates caution against interpreting the algebraic all-mode theorem as an immediate large- m sample-complexity advantage.

C. A sparse exact F_4 implementation

Set $m = n = 4$, $\ell = 0$:

$$|\Psi_{\text{cat}}\rangle = \frac{|4000\rangle + |0400\rangle + |0040\rangle + |0004\rangle}{2}. \quad (30)$$

Equation (20) makes the 25 outputs with $Q(s) \neq 0$ exactly dark. Use the two probe generators

$$H_X = X_{01} - X_{02}, \quad H_Y = Y_{01} - Y_{02}. \quad (31)$$

They are the two quadratures coupling mode 0 to the antisymmetric supermode $(|1\rangle - |2\rangle)/\sqrt{2}$.

In the column orders

$$\begin{aligned} \theta_X &= (x_{01}, x_{02}, x_{03}, x_{12}, x_{13}, x_{23}), \\ \theta_Y &= (y_{01}, y_{02}, y_{03}, y_{12}, y_{13}, y_{23}), \end{aligned} \quad (32)$$

six H_X contrasts and six H_Y contrasts have the limiting matrices

$$J_X = \begin{pmatrix} 0 & -\frac{3}{2} & 0 & 0 & -\frac{3}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{3}{4} & 0 & \frac{3}{2} \\ \frac{9}{4} & 0 & 0 & \frac{3}{4} & 0 & 0 \\ \frac{3}{4} & 0 & \frac{3}{4} & 0 & 0 & \frac{3}{2} \\ 0 & -4 & 0 & 0 & 0 & 0 \\ \frac{3}{4} & 0 & \frac{3}{2} & \frac{3}{4} & 0 & 0 \end{pmatrix}, \quad (33)$$

and

$$J_Y = \begin{pmatrix} 0 & -\frac{3}{2} & 0 & 0 & -\frac{3}{2} & 0 \\ \frac{3}{4} & 0 & 0 & \frac{3}{4} & 0 & \frac{3}{2} \\ 0 & -\frac{3}{2} & 0 & 0 & \frac{3}{2} & 0 \\ \frac{9}{4} & 0 & 0 & \frac{3}{4} & 0 & 0 \\ \frac{3}{4} & 0 & -\frac{3}{4} & 0 & 0 & \frac{3}{2} \\ \frac{3}{4} & 0 & -\frac{3}{2} & \frac{3}{4} & 0 & 0 \end{pmatrix}. \quad (34)$$

The H_X output order is

$$(0022), (0112), (0310), (1021), (1030), (1102), \quad (35)$$

and the H_Y output order is

$$(0022), (0112), (0220), (0310), (1021), (1102). \quad (36)$$

Every omitted cross block is exactly zero. The exact determinants are

$$\begin{aligned} \det J_X &= -\frac{243}{8}, & \det J_Y &= \frac{729}{32}, \\ \det \text{diag}(J_X, J_Y) &= -\frac{177147}{256}. \end{aligned} \quad (37)$$

The Gaussian-integer derivation is in Appendix C.

This sparse four-mode certificate is an implementation alternative to Theorem 3. It uses seven distinct outcomes rather than the six in the general construction, while retaining twelve scalar responses and four programmed settings because all outcomes for a probe are recorded in parallel: $\pm\epsilon H_X$ and $\pm\epsilon H_Y$. The raw limiting condition number is 7.05; row normalization gives 4.86, and ideal background-free Poisson whitening gives 4.79. These are local design diagnostics, not hardware-independent resource guarantees.

VI. FINITE PROBE ANGLE AND COUNT STATISTICS

The matrices (33)–(34) are the $\epsilon \rightarrow 0$ Jacobian of R , not exact finite-angle Jacobians. For the cat input, exact finite-angle probabilities are available from

$$A_{\mathbf{s}}(U) = \frac{1}{2} \sqrt{\frac{4!}{\prod_k s_k!}} \sum_{j=0}^3 \prod_{k=0}^3 U_{kj}^{s_k}, \quad p_{\mathbf{s}}(U) = |A_{\mathbf{s}}(U)|^2. \quad (38)$$

For the probes (31), $H^3 = 2H$, so

$$e^{i\epsilon H} = I + \frac{i \sin(\sqrt{2}\epsilon)}{\sqrt{2}} H + \frac{\cos(\sqrt{2}\epsilon) - 1}{2} H^2. \quad (39)$$

Appendix A proves that the central-contrast Jacobian has no $O(\epsilon)$ error: its first correction is $O(\epsilon^2)$. For the selected design, exact finite-angle evaluation gives the compact diagnostic in Table II. The favorable condition number at larger angle does not remove the growing model bias. Probe-only baseline subtraction in Eq. (12) is also essential: for the selected H_Y outcomes, the largest undivided baseline begins as $9\epsilon^3/8 + O(\epsilon^5)$.

For independent selected Poisson channels,

$$Y_{as} \sim \text{Pois}\{N_a[p_{as}(\boldsymbol{\theta}) + b_s]\}, \quad (40)$$

the Fisher information is

$$I_{\mu\nu}^P = \sum_{a,s} N_a \frac{\partial_\mu p_{as} \partial_\nu p_{as}}{p_{as} + b_s}. \quad (41)$$

TABLE II. Finite-angle Jacobian bias and spectral condition number in the ideal coherent model. Bias is relative Frobenius error from the limiting matrix.

ϵ	relative bias	$\kappa_2(J_\epsilon)$
0.010	7.74×10^{-4}	7.043
0.020	3.09×10^{-3}	7.033
0.050	1.92×10^{-2}	6.968
0.100	7.46×10^{-2}	6.759

At a displaced dark event, $p = c\epsilon^2 + O(\epsilon^3)$ and $\partial_\mu p = d_\mu \epsilon + O(\epsilon^2)$. The information therefore has a finite nonzero limit when $b = 0$, but is suppressed as $\epsilon^2/(c\epsilon^2 + b)$ above a fixed background floor. A multinomial model must additionally include the negative covariance between outcomes recorded in one setting. Under any stated likelihood, a Cramér–Rao bound is conditional on local unbiasedness and does not by itself give a hardware cost forecast.

This section is an ideal-model statistical layer. Its formulas and tables are reproduced by the accompanying finite-angle script and tests. Loss, partial distinguishability, source contamination, detector crosstalk, calibration drift, and optimal shot allocation require an augmented likelihood before experimental sample complexity can be claimed.

A. Cat-phase confounding

The coherent input phase is not a benign visibility parameter; it is the resource that removes the gauge. If the components of Eq. (30) acquire unknown phases α_j , infinitesimally they are generated by an input diagonal D , and

$$F_4 e^{iD} = e^{iF_4 D F_4^\dagger} F_4. \quad (42)$$

The three relative cat phases are therefore exactly confounded with the three device directions in Eq. (17). If those preparation phases are free nuisance parameters, the augmented model does not separately identify all twelve device coordinates. Independent cat-phase calibration or additional reference states are necessary. Complete dephasing returns the experiment to the number-state gauge. Even with calibrated phases, the construction is not globally injective: for a diagonal $D = \text{diag}(d_0, \dots, d_{m-1})$, the discrete conditions

$$n(d_j - d_0) \in 2\pi\mathbb{Z} \quad (1 \leq j < m) \quad (43)$$

leave the equal-phase n -photon cat unchanged up to a global phase. Consequently the corresponding finite transported transformations $F_m \exp(iD) F_m^\dagger$ are exact global aliases. They are discrete near a fixed calibrated representative and therefore do not contradict the local full-rank theorem.

VII. INTERNAL COHERENT ERRORS

Let a nominal circuit factor as

$$U_0 = U_L U_{L-1} \cdots U_1. \quad (44)$$

If a small coherent generator H_ℓ is inserted after layer ℓ , set $W_\ell = U_L \cdots U_{\ell+1}$. Then

$$U_L \cdots U_{\ell+1} e^{i\delta_\ell H_\ell} U_\ell \cdots U_1 = e^{i\delta_\ell W_\ell H_\ell W_\ell^\dagger} U_0. \quad (45)$$

For simultaneous small errors, the first-order output-equivalent generator is

$$G_{\text{eff}} = \sum_\ell \delta_\ell W_\ell H_\ell W_\ell^\dagger + O(\|\delta\|^2). \quad (46)$$

Thus the signed-probe reconstruction applies to arbitrary internal coherent, lossless errors after adjoint propagation.

This does not automatically locate a faulty component. Different layer errors can have the same output-equivalent generator, and output-diagonal projections remain outside our $m(m-1)$ -coordinate target. Layer localization requires the circuit Jacobian across additional configurations, addressable controls, or structural priors. Appendix D gives the short proof and the multi-error expansion.

VIII. RELATION TO PRIOR WORK AND LIMITATIONS

The deliberate displacement of a null measurement is a general estimation principle, not one introduced here. Girotti *et al.* established its role in resolving null-measurement non-identifiability and in asymptotically optimal pure-state estimation [8]. Our contribution is an explicit multiphoton-count implementation, the exact number-state gauge theorem, and explicit dimension-saturating local frames for a Fourier device.

Likewise, Eq. (20) is a known Fourier-cat/cyclic selection effect [4, 10]. The new role assigned to the cat is operational: it supplies the missing relative phase reference. Standard Fourier suppression experiments [1–3] and programmable photonic processors [6, 12] provide the experimental context, but preparing a phase-stable multimode multiphoton cat is substantially harder than preparing a number state. The rank-nine F_4 Fock design is the nearer-term protocol; the cat family is a complete-identifiability benchmark.

The closest comparator we found for general near-identity unitary estimation reconstructs all $d^2 - 1$ parameters and can attain the quantum Cramér–Rao bound, but uses a target qudit, two control qudits, and controlled shift and phase operations [11]. It does not provide a passive-bosonic Fourier construction from photon-counted dark outcomes. To our knowledge, prior work

has not combined Fourier-cat suppression with displaced-null readout to give an explicit all-mode, dimension-saturating local frame for coherent off-diagonal errors of a passive multiport. This is a scoped literature assessment, not a claim to the first arbitrary-unitary tomography protocol, the first displaced-null measurement, or a new Fourier-cat suppression law.

The rigorous scope is:

- local reconstruction around a calibrated nominal interferometer, not global injectivity or global unitary tomography;
- $m(m-1)$ off-diagonal coordinates only, not the $m-1$ output-diagonal $SU(m)$ directions;
- coherent lossless errors with calibrated probes and input phases;
- two probe directions and four signed settings suffice, but neither a one-probe nor a fewer-setting impossibility theorem has been proved;
- the all-mode outcome and scalar-contrast counts, and the F_4 Fock counts, are minimal only within the declared weak-probe regular-local framework; the finite-angle experiment has not been globally optimized;
- identification is local and calibrated: global aliases and off-diagonal device errors confounded with uncalibrated input phases or probe errors are not excluded;
- at $n = m$, $\alpha^2 = m^{2-m}$, so event rates decay rapidly even though the row-normalized condition bound is at most $m/(m-2)$;
- the ideal likelihood does not absorb loss or distinguishability into a universal “background” parameter [13, 14].

IX. DISCUSSION

A multiphoton dark event is not by itself a signed sensor. The known probe in Eq. (12) supplies a reference amplitude and turns photon counts into linear local observables. Which coordinates can then be reconstructed depends on both the collection of dark amplitude gradients and the phase reference carried by the input state.

This viewpoint exposes a sharp resource separation. Number states have an exact $(m-1)$ -dimensional blind subspace at F_m ; the F_4 certificate saturates the resulting rank-nine bound. A calibrated cat state lifts that continuous gauge, and the cyclic-edge construction gives an invertible $m(m-1)$ -coordinate design for every mode number. Unknown cat phases recreate precisely the obstruction the state was introduced to remove. The distinction is therefore one of identifiability, not simply better visibility.

The most immediate extensions are a joint SPAM–device likelihood, lower-cost coherent reference states, circuit-specific internal-error dictionaries, and sparse or bounded-norm probes that improve large- m count rates. Any such extension must preserve the separation between local full rank, finite-angle bias, finite-shot efficiency, and global uniqueness.

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Appendix A: Proof of the signed-probe and finite-angle theorems

Put $|\phi_e\rangle = \widehat{U}_0|\psi_e\rangle$ and suppress the event subscript. For a fixed probe $\widehat{h}$, define

$$\begin{aligned} a(t) &= \langle \mathbf{s} | e^{i\widehat{h}} | \phi \rangle, \\ b_\mu(t) &= \langle \mathbf{s} | e^{i\widehat{h}} i\widehat{G}_\mu | \phi \rangle, \\ g_\mu(t) &= \partial_{\theta_\mu} |\langle \mathbf{s} | e^{i\widehat{h}} e^{i\widehat{G}(\boldsymbol{\theta})} | \phi \rangle|^2 \Big|_{\boldsymbol{\theta}=0} \\ &= 2 \operatorname{Re}\{a(t)^* b_\mu(t)\}. \end{aligned} \quad (\text{A1})$$

Darkness gives $a(0) = 0$. Let

$$a_r = \langle \mathbf{s} | (i\widehat{h})^r | \phi \rangle, \quad b_{r\mu} = \langle \mathbf{s} | (i\widehat{h})^r i\widehat{G}_\mu | \phi \rangle. \quad (\text{A2})$$

The derivative of the response (12) is

$$\partial_{\theta_\mu} R|_0 = \frac{g_\mu(\epsilon) - g_\mu(-\epsilon)}{\epsilon}. \quad (\text{A3})$$

Taylor expansion of a and b gives

$$\begin{aligned} \partial_{\theta_\mu} R|_0 &= 4 \operatorname{Re}(a_1^* b_{0\mu}) \\ &+ 4\epsilon^2 \operatorname{Re}\left\{ \frac{a_1^* b_{2\mu}}{2} + \frac{a_2^* b_{1\mu}}{2} + \frac{a_3^* b_{0\mu}}{6} \right\} + O(\epsilon^4). \end{aligned} \quad (\text{A4})$$

Since $a_1 = \ell \mathbf{h}$ and $b_{0\mu} = \ell \mathbf{e}_\mu$, Eq. (13) follows. Writing $\ell = \mathbf{u} + i\mathbf{v}$ as a complex row, its probe row is a real linear combination of $\mathbf{u}$ and $\mathbf{v}$, which proves Eq. (14).

For a norm bound, let

$$M = \|\widehat{h}\| \leq n\|h\|, \quad L_\mu = \|\widehat{G}_\mu\| \leq n\|G_\mu\|. \quad (\text{A5})$$

Leibniz’s rule applied to Eq. (A1) yields

$$|g_\mu'''(t)| \leq 2 \sum_{k=0}^3 \binom{3}{k} M^k M^{3-k} L_\mu = 16M^3 L_\mu. \quad (\text{A6})$$

The central-difference remainder theorem therefore gives, for the convention (12),

$$|\partial_{\theta_\mu} R|_0 - 4 \operatorname{Re}(a_1^* b_{0\mu})| \leq \frac{16}{3} \epsilon^2 M^3 L_\mu. \quad (\text{A7})$$

This bound is conservative but uniform. Retaining the explicit ϵ^2 term in Eq. (A4) leaves an $O(\epsilon^4)$ remainder. Baseline subtraction does not change the derivative with respect to $\boldsymbol{\theta}$, but it removes the finite-angle probe-only offset from the observed response.

Appendix B: Fock gauge proof and exact rank-nine certificate

For $D = \operatorname{diag}(d_0, \dots, d_{m-1})$,

$$(F_m D F_m^\dagger)_{pp} = \frac{1}{m} \sum_j d_j = \frac{\operatorname{Tr} D}{m} = 0. \quad (\text{B1})$$

Conjugation commutes with exponentiation, so $e^{iK_D} F_m = F_m e^{iD}$ at the single-particle level and hence in every symmetric photon sector. A number state is an eigenvector of $\widehat{D} = \sum_j d_j \widehat{n}_j$, proving Eq. (16). Appending a known unitary V preserves the global phase, so no subsequent number measurement removes the obstruction. The map from traceless D to K_D is injective and its dimension is $m - 1$.

For completeness, the F_4 finite certificate can be evaluated with Gaussian integers. Let

$$Z_{\mathbf{r}, \mathbf{t}} = \operatorname{per}(2F_4)[\mathbf{t}, \mathbf{r}], \quad D_{\mathbf{r}, \mathbf{s}} = 2^n \sqrt{\prod_j r_j! \prod_k s_k!}. \quad (\text{B2})$$

For a dark target, the first amplitude derivatives are

$$\begin{aligned} \ell X_{pq} &= \frac{i}{D_{\mathbf{r}, \mathbf{s}}} (s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q} + s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q}), \\ \ell Y_{pq} &= \frac{1}{D_{\mathbf{r}, \mathbf{s}}} (s_p Z_{\mathbf{r}, \mathbf{s} - \mathbf{e}_p + \mathbf{e}_q} - s_q Z_{\mathbf{r}, \mathbf{s} + \mathbf{e}_p - \mathbf{e}_q}). \end{aligned} \quad (\text{B3})$$

Negative occupations are omitted. These follow directly from the Fock ladder factors; the square roots cancel against the changed-state normalization.

Substitution into Eq. (13), for the row order of Table I, gives

$$J_F = \begin{pmatrix} 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & 0 & 0 & -\frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & 0 & 0 & \frac{3}{2} \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & -\frac{3}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{2} & 0 & 0 & 0 & -\frac{3}{2} \\ -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 2 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \quad (\text{B4})$$

Each row annihilates the coordinate vectors of K_A, K_B, K_C . The minor in columns

$$(x_{01}, y_{01}, x_{02}, y_{02}, x_{03}, y_{03}, x_{12}, y_{12}, y_{13}) \quad (\text{B5})$$

has determinant 81/8. Hence $\text{rank } J_F = 9$ and the three independent gauge vectors form its complete nullspace.

Appendix C: Fourier-cat frame proof and sparse rank-twelve certificate

For the component $|ne_j\rangle$, every term in the repeated-column permanent has the same phase. Equation (1) gives

$$\mathcal{A}_{|ne_j\rangle, \mathbf{s}}(F_m) = \sqrt{\frac{n!}{\prod_k s_k!}} m^{-n/2} \omega^{jQ(\mathbf{s})}. \quad (\text{C1})$$

Multiplying by the coefficients in Eq. (19) and summing over j gives

$$\frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \omega^{j[Q(\mathbf{s})-\ell]} = \sqrt{m} \delta_{Q(\mathbf{s}), \ell}, \quad (\text{C2})$$

which proves Eq. (20).

We next derive Eq. (22) and complete the all-mode proof. For a general single-particle matrix U , the cat amplitude is the polynomial

$$A_{\mathbf{s}}(U) = \sqrt{\frac{n!}{\prod_a s_a!}} \frac{1}{\sqrt{m}} \sum_{j=0}^{m-1} \prod_{a=0}^{m-1} U_{aj}^{s_a}. \quad (\text{C3})$$

Along $U(\theta) = e^{i\theta H} F_m$, $\dot{U}_{aj} = i \sum_b H_{ab} F_m(b, j)$. Differentiating Eq. (C3) at zero produces terms proportional to

$$\sum_{j=0}^{m-1} \omega^{j[Q(\mathbf{s})+b-a]} = m \delta_{b, a-c} \quad (\text{C4})$$

when $Q(\mathbf{s}) = c$. Collecting the factorial and Fourier normalizations gives

$$\left. \frac{d}{d\theta} A_{\mathbf{s}}(e^{i\theta H} F_m) \right|_{\theta=0} = i\alpha_{\mathbf{s}} \sum_a s_a H_{a, a-c} = iL_{\mathbf{s}}(H), \quad (\text{C5})$$

as asserted.

For the selected outcome (23), all factorials are the same and $\alpha = \sqrt{n} m^{(1-n)/2}$. In a non-antipodal cyclic class, the m complex coordinates $z_a = H_{a, a-c}$ are independent, because their Hermitian conjugates belong to the unselected class $-c$. The selected functionals form the complex matrix $(n-1)I + P_c$. If it annihilates z , unitarity of the permutation gives

$$(n-1)\|z\| = \|P_c z\| = \|z\|, \quad (\text{C6})$$

so $z = 0$ for $n > 2$. In the self-conjugate antipodal class, Hermiticity instead reduces each edge to $z_p = x_p + iy_p$, and its functional is $nx_p + i(n-2)y_p$. This is invertible over the reals. Each undirected off-diagonal edge appears in exactly one selected cyclic class, so these blocks are a direct-sum decomposition of $\mathcal{H}_{\text{off}}(m)$.

It remains to show that the complex functionals are observable from counts. For the probes defined in Eq. (25), direct substitution gives Eq. (26). Since $\ell = iL$, Theorem 1 reduces here to

$$DR_{\mathbf{s}, h}(0)[H] = 4 \text{Re}\{L_{\mathbf{s}}(h)^* L_{\mathbf{s}}(H)\}. \quad (\text{C7})$$

The R and T rows are therefore nonzero multiples of the real and imaginary parts of every $L_{\mathbf{s}}$. The preceding direct-sum argument proves full rank. There are exactly $m(m-1)$ such real rows. Since Corollary 1 bounds one event by two rows, the event and scalar-response counts meet the weak-probe regular-local lower bounds. Analytic dependence on ϵ and persistence of a nonzero determinant then establish the finite-angle local statement through the ordinary inverse-function theorem.

The conditioning claim follows from the singular values $|n-1+\lambda|$, $|\lambda| = 1$, of each non-antipodal block, and from the two gains $n, n-2$ in an antipodal block. Their ratio is at most $n/(n-2)$ after known reference-row normalization. Finally, $\alpha^2 = nm^{1-n}$, which becomes Eq. (29) when $n = m$.

We now certify the sparse F_4 matrices without numerical zero tests. For any four-mode unitary,

$$A_{\mathbf{s}}(U) = \frac{1}{2} \sqrt{\frac{4!}{\prod_k s_k!}} \sum_{j=0}^3 \prod_{k=0}^3 U_{kj}^{s_k}. \quad (\text{C8})$$

Its derivative in the direction $\dot{U}_\mu = iG_\mu F_4$ is

$$\dot{A}_{s,\mu} = \frac{1}{2} \sqrt{\frac{4!}{\prod_k s_k!}} \sum_{j=0}^3 \sum_{k:s_k>0} s_k (\dot{U}_\mu)_{kj} U_{kj}^{s_k-1} \prod_{\substack{r=0 \\ r \neq k}}^3 U_{rj}^{s_r}. \quad (C9)$$

At $U = F_4$, all entries lie in $\mathbb{Q}(i)$ times the displayed normalization. The limiting physical response is

$$J_{s,\mu} = 4 \operatorname{Re} \left(\dot{A}_{s,h}^* \dot{A}_{s,\mu} \right). \quad (C10)$$

Evaluating Eq. (C9) for the outputs (35)–(36) gives Eqs. (33)–(34). Direct expansion gives

$$\det J_X = -243/8, \quad \det J_Y = 729/32. \quad (C11)$$

The H_X rows have zero entries in all six Y columns and the H_Y rows have zero entries in all six X columns. Thus the full determinant is their product, Eq. (37). This is

an exact rational certificate. The released verification independently constructs the same rows from Gaussian-integer permanent moments and asserts every cross-block zero.

Appendix D: Internal-error propagation

Write $U_0 = W_\ell V_\ell$, with $W_\ell = U_L \cdots U_{\ell+1}$ and $V_\ell = U_\ell \cdots U_1$. Then

$$\begin{aligned} W_\ell e^{i\delta H_\ell} V_\ell &= W_\ell e^{i\delta H_\ell} W_\ell^\dagger U_0 \\ &= e^{i\delta W_\ell H_\ell W_\ell^\dagger} U_0, \end{aligned} \quad (D1)$$

which proves Eq. (45) exactly for one inserted error. For several errors, expanding each exponential to first order and collecting terms gives Eq. (46). Commutators enter at second order, so the layer-to-output dictionary is linear only in the local regime.

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